Week 6: The EM-Algorithm

MATH-517 Statistical Computation and Visualization

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EM Algorithm - Recap

- $\mathbf{X}_{obs}$ are the **observed** random variables
- $\mathbf{X}_{miss}$ are the **missing** random variables
- $\ell_{comp}(\theta)$ is the **complete** log-likelihood of $\mathbf{X} = (\mathbf{X}_{obs}, \mathbf{X}_{miss})$
  - maximizing this to obtain MLE is supposed to be *simple*
  - θ denotes all the parameters, e.g., contains μ and Σ

Our task is to maximize $\ell_{obs}(\theta)$, the **observed** log-likelihood of $\mathbf{X}_{obs}$

EM Algorithm: Start from an initial estimate $\theta^{(0)}$ and for $l = 1, 2, ...$ iterate the following two steps until convergence:

- **E-step**: calculate $\mathbb{E}_{\hat{\theta}^{(l-1)}}\left[\ell_{comp}(\theta) \middle| \mathbf{X}_{obs} = \mathbf{x}_{obs}\right] =: Q\left(\theta, \theta^{(l-1)}\right)$
- **M-step**: optimize $\arg\max_{\theta} Q\left(\theta, \theta^{(l-1)}\right) =: \theta^{(l)}$

Section 1

Some Properties of EM

Monotone Convergence

Proposition 1: $\ell_{obs}(\theta^{(l)}) \geq \ell_{obs}(\theta^{(l-1)})$

- a step of the EM algorithm will never decrease the objective value
- algorithms with this property are typically
  - numerically stable (good)
  - convergent under mild conditions (good)
- the algorithm is guaranteed to converge to a stationary point of the likelihood under a continuity condition on $Q(\cdot, \cdot)$; see Theorem 3.2 in McLachlan and Krishnan, 2007
  - convergence to a unique MLE requires unimodality of the likelihood (among other conditions)
  - prone to get stuck in local maxima (bad)

Monotone Convergence - Proof

The joint density for the complete data $\mathbf{X} = (\mathbf{X}_{obs}, \mathbf{X}_{miss})^\top$ satisfies $f_\theta(\mathbf{X}) = f_\theta(\mathbf{X}_{miss}|\mathbf{X}_{obs}) f_\theta(\mathbf{X}_{obs})$ and hence

$$\ell_{comp}(\theta) = \log f_\theta(\mathbf{X}_{miss}|\mathbf{X}_{obs}) + \ell_{obs}(\theta)$$

Notice that $\ell_{obs}(\theta)$ does not depend on $\mathbf{X}_{miss}$ and hence we can condition on $\mathbf{X}_{obs}$ under any value of the parameter θ without really doing anything:

$$\begin{aligned}
\ell_{obs}(\theta) &= \mathbb{E}_{\theta^{(l-1)}}\left\{ \ell_{comp}(\theta) - \log f_\theta(\mathbf{X}_{miss}|\mathbf{X}_{obs}) \middle| X_{obs} \right\} \\
&= \underbrace{\mathbb{E}_{\theta^{(l-1)}}\left\{ \ell_{comp}(\theta) \middle| X_{obs} \right\}}_{=Q\left(\theta, \theta^{(l-1)}\right)} - \underbrace{\mathbb{E}_{\theta^{(l-1)}}\left\{ \log f_\theta(X_{miss}|X_{obs}) \middle| X_{obs} \right\}}_{=:H\left(\theta, \theta^{(l-1)}\right)}
\end{aligned}$$

Thus, when we take $\hat{\theta}^{(l)} = \arg\max_\theta Q(\theta, \hat{\theta}^{(l-1)})$, we only have to show that we have not increased $H(\cdot, \theta^{(l-1)})$

Monotone Convergence - Proof

Dividing and multiplying by $f_{\theta^{(l-1)}}(X_{miss}|X_{obs})$ and using Jensen's inequality, we obtain just that:

$$H(\theta,\theta^{(l-1)}) = \mathbb{E}_{\theta^{(l-1)}}\left\{ \ln \frac{f_\theta(X_{miss}|X_{obs})}{f_{\theta^{(l-1)}}(X_{miss}|X_{obs})} \Big| X_{obs} \right\} + H(\theta^{(l-1)},\theta^{(l-1)})$$

$$\leq \ln \underbrace{\mathbb{E}_{\theta^{(l-1)}}\left\{ \frac{f_\theta(X_{miss}|X_{obs})}{f_{\theta^{(l-1)}}(X_{miss}|X_{obs})} \Big| X_{obs} \right\}}_{=\int \frac{f_\theta(x_{miss}|X_{obs})}{f_{\theta^{(l-1)}}(x_{miss}|X_{obs})} f_{\theta^{(l-1)}}(x_{miss}|X_{obs}) dx_{miss} = 1} + H(\theta^{(l-1)},\theta^{(l-1)})$$

and so indeed $H(\theta,\theta^{(l-1)}) \leq H(\theta^{(l-1)},\theta^{(l-1)})$

Speed of Convergence: Definition

We have an iterative algorithm that is trying to find the maximum/minimum of a function and we want to estimate how long it will take to reach that optimal value

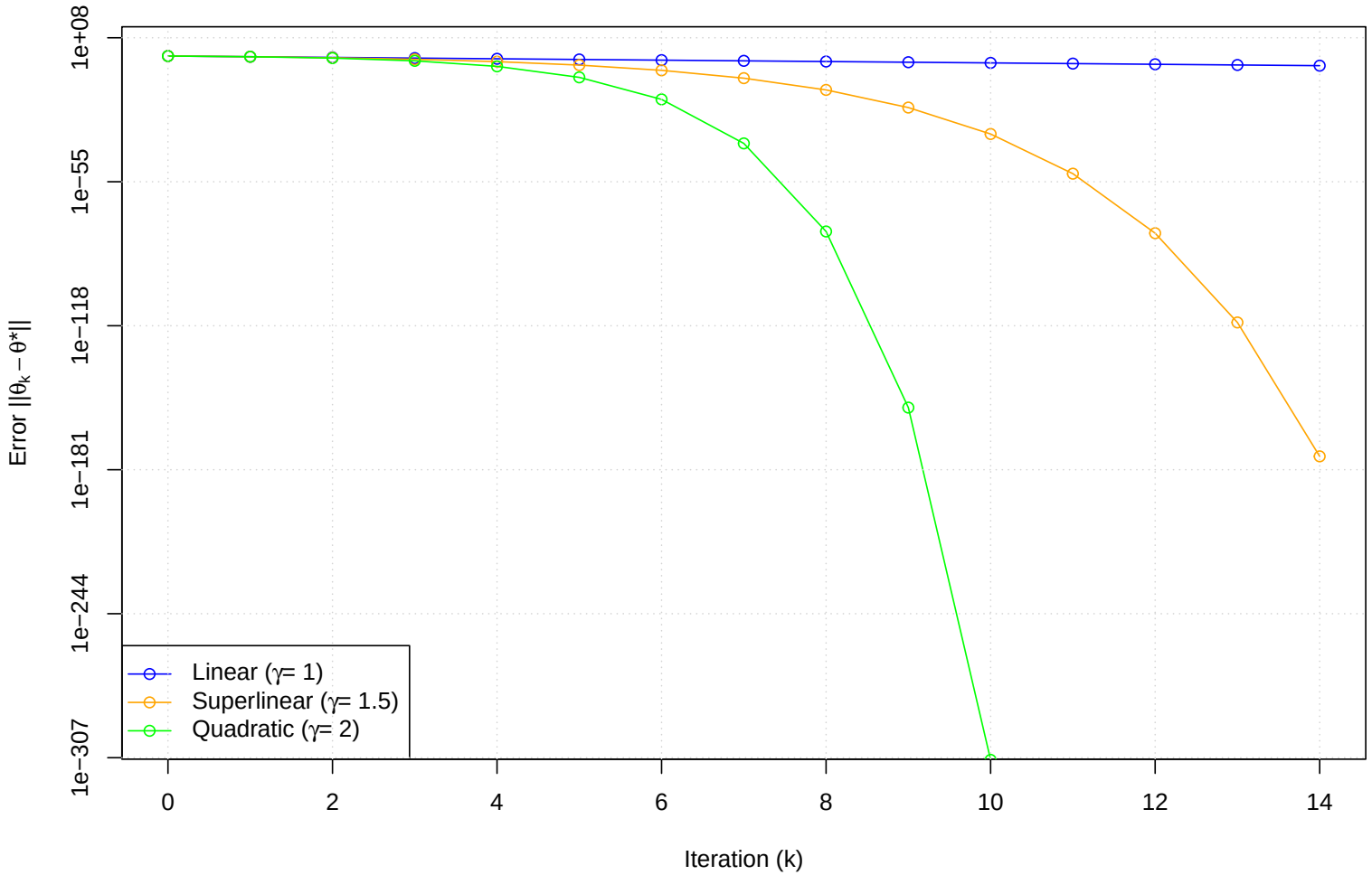
For an iterative algorithm that converges to a solution $\theta^\star$, if there is a real number γ and a constant integer k_0, such that for all $k > k_0$, we have

$$\left\|\theta^{(k+1)} - \theta^\star\right\| \leq q \left\|\theta^{(k)} - \theta^\star\right\|^\gamma$$

with q being a positive constant independent of k, then we say that the algorithm has a convergence rate of order γ. An algorithm has

- first-order or linear convergence if $\gamma = 1$ and $q \in (0, 1)$ (sublinear if $q = 1$)
- superlinear convergence if $1 < \gamma < 2$ (quasi-Newton methods)
- second-order or quadratic convergence if $\gamma = 2$ (Newton)

Speed of Convergence



Speed of Convergence for EM

Consider the iteration mapping $M : \theta^{(l-1)} \mapsto \theta^{(l)}$, assumed continuous

- if $\theta^{(l)} \to \theta^\star$ as $l \to \infty$, then it must be a fixed point: $M(\theta^\star) = \theta^\star$
- in a neighborhood of $\theta^\star$, a 1st order Taylor expansion:

$$\theta^{(l+1)} = M(\theta^{(l)}) \approx \theta^\star + \left.\frac{\partial M(\theta)}{\partial \theta^\top}\right|_{\theta=\theta^\star} (\theta^{(l)} - \theta^\star)$$

  yields

$$\theta^{(l+1)} - \theta^\star \approx \mathbf{J}(\theta^\star)\,(\theta^{(l)} - \theta^\star),$$

  where $\mathbf{J}(\cdot)$ is the Jacobian of M
- Rate is linear as $\|\theta^{(l)} - \theta^\star\| \approx \|\mathbf{J}(\theta^\star)\|^l\, \|\theta^{(0)} - \theta^\star\|$, for any norm on $\mathbb{R}^d$
- Global rate of convergence $\lim \|\theta^{(l+1)} - \theta^{(l)}\| / \|\theta^{(l)} - \theta^{(l-1)}\| = \rho\{\mathbf{J}(\theta^\star)\} \equiv$ largest eigenvalue of $\mathbf{J}(\theta^\star)$, under regularity conditions
- Smaller $\rho\{\mathbf{J}(\theta^\star)\}$ means faster global convergence
- If $\rho\{\mathbf{J}(\theta^\star)\} < 1$, then M is a contraction and we may hope for convergence

Speed of Convergence for EM

It can be shown that:

$$\mathbf{J}(\theta^\star) = \mathbf{J}_{comp}^{-1}(\theta^\star)\, \mathbf{J}_{miss}(\theta^\star),$$

where $\mathbf{J}_{comp}$ and $\mathbf{J}_{miss}$ are Fisher information of the complete resp. missing data

Then, $\rho\{\mathbf{J}(\theta^\star)\}$ is the largest eigenvalue of the information ratio matrix measuring the fraction of missing information about θ

$\Rightarrow$ the bigger the proportion of missing information, the slower the convergence

Exponential Families

Let the density of the complete data be from the exponential family, i.e.,

$$f_X(\mathbf{x}) = \exp\left\{\eta(\theta)^\top \mathbf{T}(\mathbf{x}) - g(\theta)\right\} h(\mathbf{x})$$

where

- $\theta \in \Theta \subset \mathbb{R}^p$
- $\mathbf{T}(\mathbf{x}) = \left(T_1(\mathbf{x}), \ldots, T_q(\mathbf{x})\right)^\top$ is the *sufficient statistic* for θ
- $\eta : \mathbb{R}^p \to \mathbb{R}^p$, $g : \mathbb{R}^p \to \mathbb{R}$ and $h : \mathbb{R}^d \to \mathbb{R}$

Assuming $\eta(\theta) = \theta$, i.e., θ is the canonical parameter, we have

$$\ell_{comp}(\theta) = \sum_{n=1}^{N} \theta^\top \mathbf{T}(\mathbf{X}_n) + \ln h(\mathbf{X}_n) - N g(\theta)$$

and

$$Q(\theta, \theta^{(l-1)}) = \sum_{n=1}^{N} \theta^\top \mathbf{t}_n^{(l)} + \mathbb{E}_{\theta^{(l-1)}}\left[\ln h(\mathbf{X}_n) \middle| \mathbf{X}_{obs}\right] - N g(\theta),$$

where $\mathbf{t}_n^{(l)} = \mathbb{E}_{\theta^{(l-1)}}\left[T(\mathbf{X}_n) \middle| \mathbf{X}_{obs}\right]$

Exponential Families

- It is straightforward that for the E-step we will only need to compute the conditional expectations of the complete-data sufficient statistics

$$\mathbb{E}_{\theta^{(l-1)}}\left[T_i(\mathbf{X})\middle|\mathbf{X}_{obs}\right], \quad i = 1, \dots, p$$

- The M-step is equivalent to finding the expressions for the complete-data ML estimates of θ and replacing the complete-data sufficient statistics in these expressions with their conditional expectations computed in the E-step
- This is equivalent to solving:

$$\frac{\partial Q}{\partial \theta} = 0 \Rightarrow \frac{\partial g(\theta)}{\partial \theta} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{t}_n^{(l)}$$

Since $\frac{\partial g(\theta)}{\partial \theta} = \mathbb{E}_\theta[\mathbf{T}(\mathbf{X})]$, the M-step is equivalent to finding θ that satisfies

$$\mathbb{E}_\theta[\mathbf{T}(\mathbf{X})] = \frac{1}{N}\sum_{n=1}^{N} \mathbf{t}_n^{(l)}$$

Note: This applies, e.g., to Example 3 from Week 5

Section 2

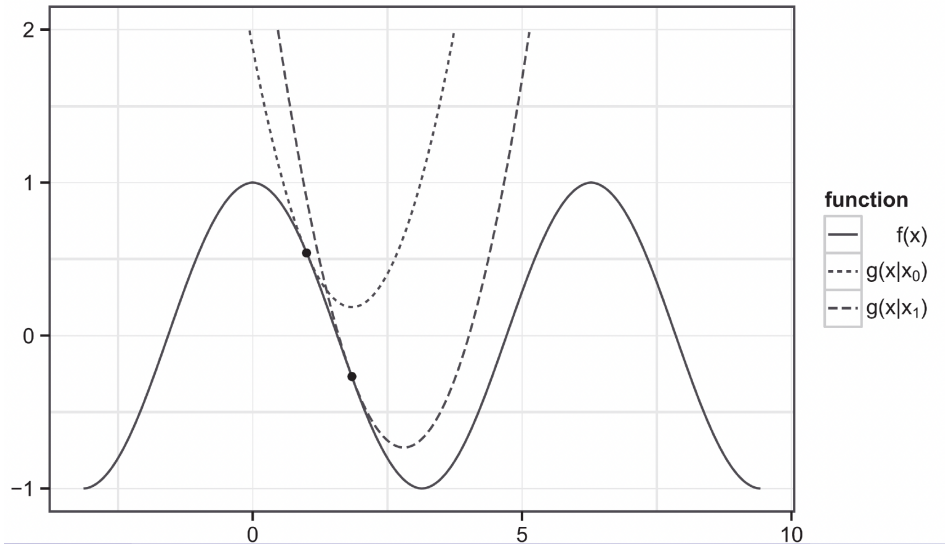
MM Algorithms

MM Algorithms

Definition: A function $g(\mathbf{x} \mid \mathbf{x}^{(l)})$ is said to **majorize** a function $f : \mathbb{R}^p \to \mathbb{R}$ at $\mathbf{x}^{(l)}$ provided

- $f(\mathbf{x}) \leq g(\mathbf{x}|\mathbf{x}^{(l)}), \quad \forall\, \mathbf{x}$
- $f(\mathbf{x}^{(l)}) = g(\mathbf{x}^{(l)}|\mathbf{x}^{(l)})$

In other words, the surface $\mathbf{x} \mapsto g(\mathbf{x}|\mathbf{x}^{(l)})$ is above the surface $f(\mathbf{x})$, and it is touching it at $\mathbf{x}^{(l)}$



MM Algorithms

Assume our goal is to minimize a function $f : \mathbb{R}^p \to \mathbb{R}$

The basic idea of the MM algorithm is to start from an initial guess $\mathbf{x}^{(0)}$ and for $l = 1, 2, ...$ iterate between the following steps until convergence:

- **Majorization step**: construct $g(\mathbf{x}|\mathbf{x}^{(l-1)})$, i.e., construct a majorizing function to f at $\mathbf{x}^{(l-1)}$
- **Minimization step**: set $\mathbf{x}^{(l)} = \arg\min_{\mathbf{x}} g(\mathbf{x}|\mathbf{x}^{(l-1)})$, i.e., minimize the majorizing function

$\to$ MM stands for "Majorization-Minimization" or "Minorization-Maximization"

Monotone convergence property is trivially guaranteed by construction:

$$f(\mathbf{x}^{(l)}) \leq g(\mathbf{x}^{(l)}|\mathbf{x}^{(l-1)}) \leq g(\mathbf{x}^{(l-1)}|\mathbf{x}^{(l-1)}) = f(\mathbf{x}^{(l-1)})$$

E-step Minorizes

With extra minus sign, the EM is:

$$\textbf{E-step:} \quad Q(\theta|\theta^{(l-1)}) := \mathbb{E}_{\theta^{(l-1)}}\big[-\ell_{comp}(\theta)\big|X_{obs}\big]$$
$$\textbf{M-step:} \quad \theta^{(l)} := \arg\min_{\theta} Q(\theta|\theta^{(l-1)})$$

From the proof of Proposition 1 above, we have (with the extra sign)

$$-\ell_{obs}(\theta) = -Q(\theta|\theta^{(l-1)}) + H(\theta, \theta^{(l-1)})$$

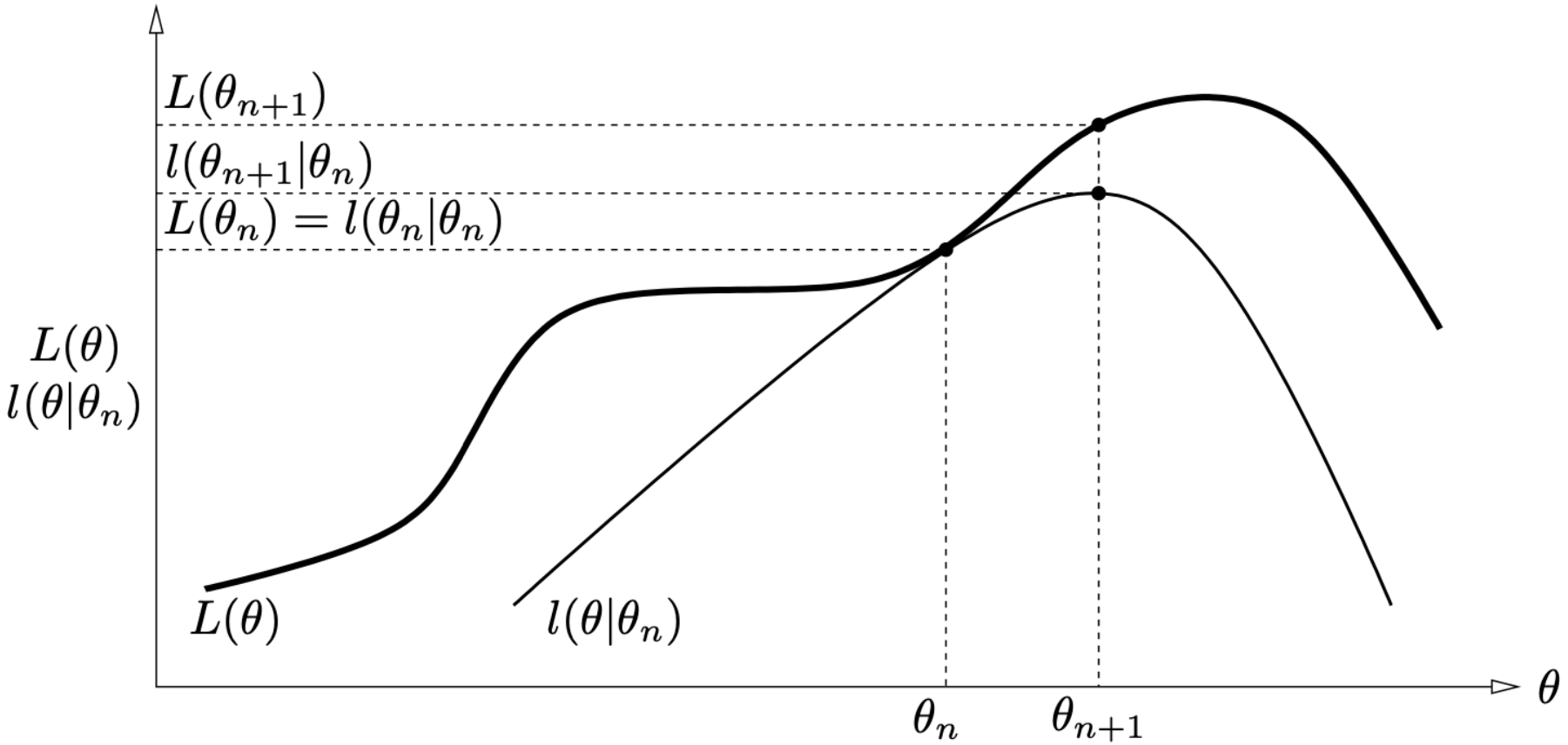
and since $H(\theta, \theta^{(l-1)}) \leq H(\theta^{(l-1)}, \theta^{(l-1)})$, we obtain

$$-\ell_{obs}(\theta) \leq -Q(\theta|\theta^{(l-1)}) + H(\theta^{(l-1)}, \theta^{(l-1)}) =: \widetilde{Q}(\theta|\theta^{(l-1)})$$

with equality at $\theta = \theta^{(l-1)}$

- $\widetilde{Q}(\theta|\theta^{(l-1)})$ is majorizing $-\ell_{obs}(\theta)$ at $\theta = \theta^{(l-1)}$
- $H(\theta^{(l-1)}, \theta^{(l-1)})$ is a constant (w.r.t. θ)

Graphical interpretation revisited

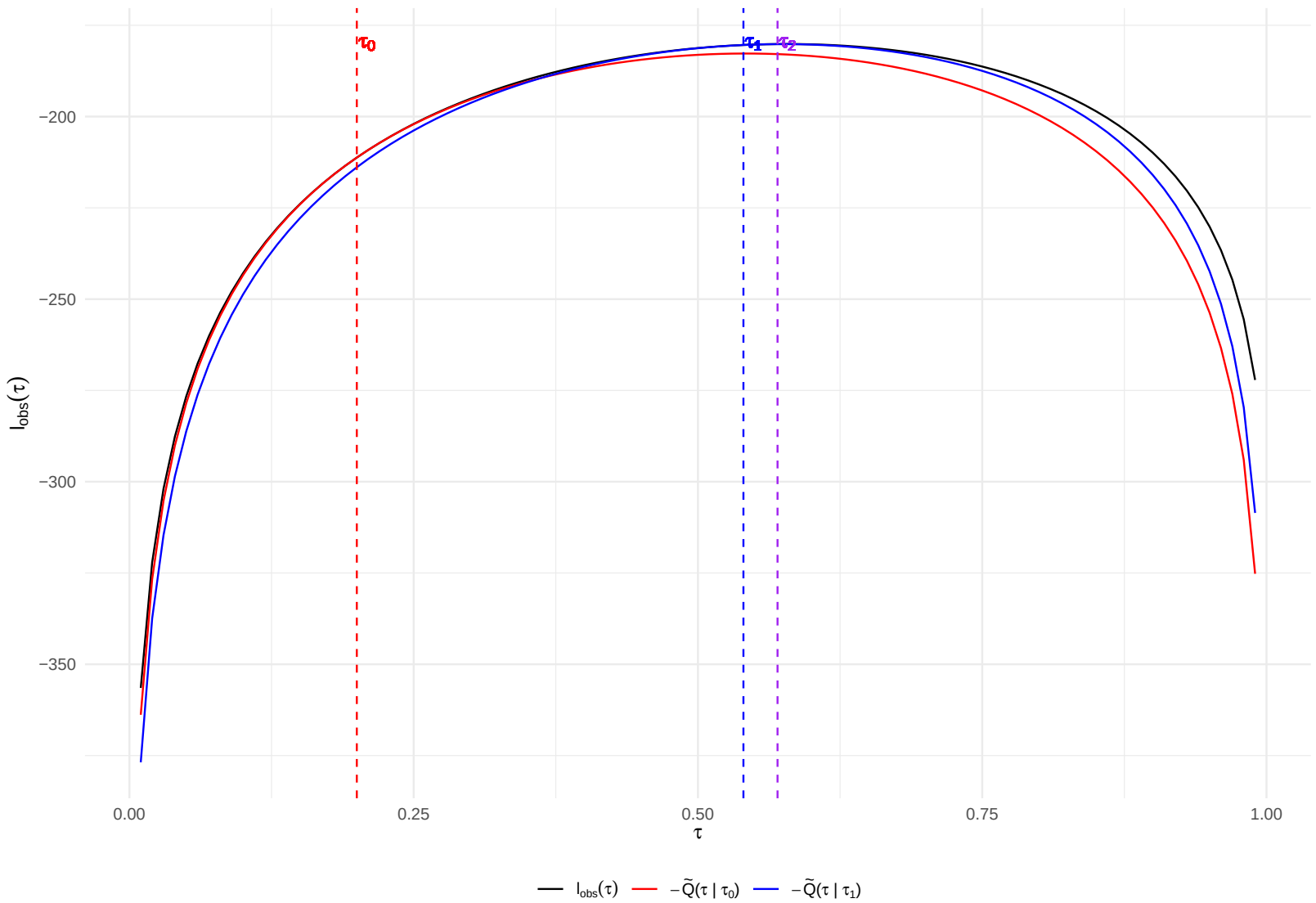


$$\ell(\theta \mid \theta_n) = -\tilde{Q}(\theta \mid \theta_n) = Q(\theta|\theta^{(n)}) - Q(\theta^{(n)}|\theta^{(n)}) + \ell_{obs}(\theta^{(n)}) \leq \ell_{obs}(\theta) = L(\theta)$$

Example 2 (Week 5) revisited

```
rmixnorm <- function(N, tau, mu1=3, mu2=0, sigma1=0.5, sigma2=1){
  ind <- I(runif(N) > tau)
  X <- rep(0,N)
  X[ind] <- rnorm(sum(ind), mu1, sigma1)
  X[!ind] <- rnorm(sum(!ind), mu2, sigma2)
  return(X)
}
dmixnorm <- function(x, tau, mu1=3, mu2=0, sigma1=0.5, sigma2=1){
  y <- (1-tau)*dnorm(x,mu1,sigma1) + tau*dnorm(x,mu2,sigma2)
  return(y)
}
ell_obs <- function(X, tau, mu1=3, mu2=0, sigma1=0.5, sigma2=1){
  return(sum(log(dmixnorm(X, tau, mu1, mu2, sigma1, sigma2))))
}
Q <- function(t, tl){
  gammas <- dnorm(X)*tl/dmixnorm(X, tl)
  qs <- dnorm(X,3,0.5)^(1-gammas)*dnorm(X)^gammas*t^gammas*(1-t)^(1-gammas)
  return(sum(log(qs)))
}
```

Two Steps Visualized



MM example: Finding a (sample) median

Consider the sequence of observations $x_1, \ldots, x_N$. The sample median θ minimizes the non-differentiable criterion

$$f(\theta) = \sum_{n=1}^{N} |x_n - \theta|$$

The quadratic function

$$h_n\left(\theta \mid \theta^l\right) = \frac{1}{2}\frac{\left(x_n - \theta\right)^2}{\left|x_n - \theta^l\right|} + \frac{1}{2}\left|x_n - \theta^l\right|$$

majorizes $|x_n - \theta|$ at $\theta^l \Rightarrow g\left(\theta \mid \theta^l\right) = \sum_{n=1}^{N} h_n\left(\theta \mid \theta^l\right)$ majorizes $f(\theta)$

The minimum of $g\left(\theta \mid \theta^l\right)$ occurs at $\theta^{l+1} = (\sum_{n=1}^{N} w_n^l x_n)/(\sum_{n=1}^{N} w_n^l)$, for $w_n^l = \left|x_n - \theta^l\right|^{-1}$

$\rightarrow$ generalizes to L_1 regression and quantile regression

MM Convergence

Theorem. (Lange, 2013, Proposition 12.4.4)
Suppose that all stationary points of $f(\mathbf{x})$ are isolated and that the *differentiability*, *coerciveness*, and *convexity* assumptions are true. Then any sequence that iterates $\mathbf{x}^{(l)} = M(\mathbf{x}^{(l-1)})$, generated by the iteration map $M(\cdot)$ of the MM algorithm, possesses a limit, and that limit is a stationary point of $f(\mathbf{x})$. If $f(\mathbf{x})$ is strictly convex, then $\lim_{l\to\infty} \mathbf{x}^{(l)}$ is the minimum point.

- *differentiability* - conditions on majorizations guaranteeing differentiability of the iteration map M
- *coerciveness* - upper level sets of f $\{\mathbf{x} : f(\mathbf{x}) \leq f(\mathbf{x}_0)\}$ are compact. $f(x) \to \infty$ for $\|x\| \to \infty$ prevents a minimizing sequence from escaping to infinity
- *convexity* - Without it, we would say that all limit points (which however might not exist without convexity) are stationary points and MM converges to one of them. Under convexity, the stationary point is the unique global minimizer

MM Remarks

- MM algorithms can linearize an optimization problem (mixture of Gaussians)
- MM algorithms can turn a non-differentiable problem into a smooth problem
- The rate of convergence depends on how well the majorizer/minorizer $g(\mathbf{x} \mid \mathbf{x}^{(l)})$ approximates the target $f(\mathbf{x})$
- There exist methods for accelerating the convergence of MM and EM algorithms (e.g., Aitken's method); see Zhou et al. (2009) and Chapter 4 in McLachlan and Krishnan (2008)

Concluding EM Remarks

- EM is just MM with majorization achieved by Jensen's inequality
- due to the monotone convergence property of all MM algorithms, EM
  - is numerically stable (obs llik improves monotonically)
  - typically converges
  - but can get stuck in a local minimum/maximum

How to choose starting parameters in mixture of Gaussian?

Hastie and Tibshirani (Elements of Statistical Learning, pg. 293) recommend constructing initial guesses as follows:

- For $\hat{\mu_1}$ and $\hat{\mu_2}$, randomly select two y_i values
- For $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$, set both equal to the overall sample variance $\sum_{i=1}^{N} (y_i - \bar{y}) (y_i - \bar{y})^T / N$
- For $\hat{\tau}$, begin at 0.50

In practice, the EM algorithm is often run using several different combinations of starting parameter estimates $\Rightarrow$ prevents relying on one set of starting parameters that may get stuck in a local max

Concluding EM Remarks

- EM computational costs per iteration are typically favorable (simple steps), but
- convergence relatively slow (many steps)
  - linear in the neighborhood of the limit
  - in practice monitored by looking at $\|\mathbf{x}^{(l)} - \mathbf{x}^{(l-1)}\|$ ($\|\theta^{(l)} - \theta^{(l-1)}\|$) and $|f(\mathbf{x}^{(l)}) - f(\mathbf{x}^{(l-1)})|$ ($|\ell_{obs}(\theta^{(l)}) - \ell_{obs}(\theta^{(l-1)})|$)
- the M-step may not have a closed form solution, but is typically much simpler than the original problem (e.g., yielding separable optimization subproblems)
  - if inner iteration for the M-step, early stopping is often desirable (e.g., Generalized EM, where the M-step only needs to increase the Q function)
  - ex.: logistic regression with missing covariates (M-step solved by IRLS)

References

- Lange, K. (2013). *Optimization*. 2nd Edition.
- Lange, K. (2016). *MM optimization algorithms*.
- McLachlan, G.J., & Krishan, T. (2008). *The EM algorithm and extensions*.

Main Project

Go to Main project for details